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RESEARCH

Improving knowledge graphs via data-based causal structures

Derui Lyu[1] · Xiangyu Wang[1] · Lyuzhou Chen[1] · Taiyu Ban[1] · Kaiming Xiao[2] · Lihua Liu[2]

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Abstract
Knowledge graphs, prevalent in a multitude of domains, have gained significant traction over the years. However, these graphs, often created from noisy sources using imperfect methods, suffer from inaccuracies, inconsistencies, and incompleteness. The imperative to rectify and supplement these issues is frequently addressed by costly domain-expert-based strategies or learning-based techniques, which can prove unreliable in noisy environments. Real-world data, with its inherent objectivity and implicit knowledge, can serve as an external source of knowledge. The ready availability of big data today presents an opportunity for improving knowledge graphs. Yet, the challenge of extracting latent information from such data has often deterred the inclusion of numerical data in current research. This paper introduces a pioneering approach that leverages external numerical data to enhance knowledge graph quality. The proposed method commences by mining statistical relationships, such as correlation and causality from the data. Following a thorough analysis of causal patterns and an evaluation of relation strengths, it identifies and eliminates redundant knowledge. The empirical results from numerous experiments validate the efficacy of our approach. The proposed method outperforms existing learning-based methods, demonstrating superior accuracy and stability in knowledge correction.

✉ Lihua Liu
gogonudt@126.com

Derui Lyu
drlv@mail.ustc.edu.cn

Xiangyu Wang
sa312@ustc.edu.cn

Lyuzhou Chen
clz31415@mail.ustc.edu.cn

Taiyu Ban
banty@mail.ustc.edu.cn

Kaiming Xiao
kmxiao@nudt.edu.cn

[1] School of Computer Science and Technology, University of Science and Technology of China, Hefei 230026, China

[2] Laboratory for Big Data and Decision, National University of Defence Technology, Changsha 410003, China

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1 Introduction

As knowledge graphs (KGs) evolve, many methods based on KGs are being used in various professional fields, including business analytics [1], predictive maintenance [2], and drug discovery [3], among others [4, 5]. Despite their widespread adoption, KGs, primarily generated from noisy data sources and through automated or semi-automated processes, are susceptible to the infiltration of inaccurate knowledge [6]. This compromises their quality and effectiveness [7]. For instance, in drug discovery, such inaccuracies could culminate in unanticipated side effects during treatment [8], underscoring the critical need for methodologies to enhance the integrity of KGs.

In the professional domain, numerous strategies have been proposed for KG refinement, encompassing expert-based [9] and learning-based methodologies [10]. Nevertheless, these approaches are not without their challenges: (1) Expert-based correction methods are accurate but require a lot of work and can be expensive [9]. They depend on manual work and can vary because different experts have different opinions [11]. (2) Learning-based automatic correction methods using learning algorithms do not need experts, but they can be affected by errors in the data. These methods are sensitive to small mistakes, which are hard to check and fix [12]. (3) In many professional areas, especially in cutting-edge fields, knowledge is always changing, or it is hard to verify. This means that experts are needed to make sense of the raw data, which then needs to be checked. This highlights the need for an automated, unbiased, and effective way to correct KGs. The solution should be designed specifically for the needs of the particular field and its applications.

Data, which is concrete representation of real-world phenomena, naturally hold the power to reveal new information, and in many cases, it serves as a direct source of human understanding [13]. In professional contexts, numerical data are plentiful and readily accessible, spanning operational, sales, and experimental data [14]. As an external information source, these data contain complex patterns, including variable correlations and distributions. These patterns can be seen as latent or not clearly defined knowledge [15]. Data's relative objectivity, stemming from its lack of human bias, renders it a promising resource for assisting KG correction [16]. For instance, examining clinical data to learn whether aspirin is helpful for high blood pressure could greatly simplify the process of verifying knowledge [17], as shown in (Figs. 1). Nevertheless, the different characteristics of numerical data and knowledge presents a challenge for their concurrent utilization. This situation requires a special approach that helps in confirming accurate knowledge and rejecting incorrect information within knowledge graphs.

Investigating the connections or cause-and-effect relationships among attributes of entities is a key method for extracting knowledge from data. This process can be formalized using statistical techniques like causal discovery and testing for conditional independence. These methods allow for the automatic extraction of detailed semantic relationships from data [18]. By quantitatively representing these statistical patterns, we can employ data for knowledge supervision. Drawing inspiration from this, our paper introduces an innovative method for KG correction, leveraging correlation and causal patterns discerned from data. This method first measures the strength of the relationship, showing how much two variables depend on each other and indicating the chance of a relationship within the KG. Then, it uncovers causal relationships from the data and represents them in a causal graph, which is used to adjust

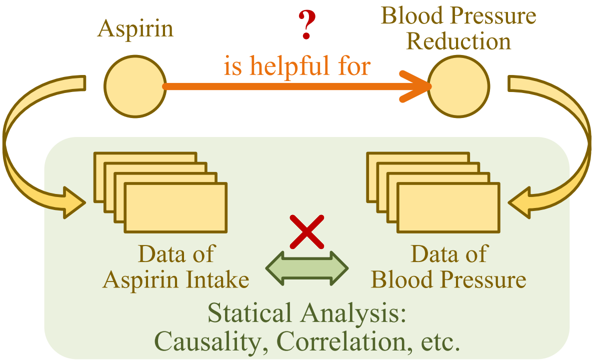


Fig. 1 An example of whether aspirin is helpful for blood pressure reduction can be judged with a statistical analysis between aspirin intake and blood pressure

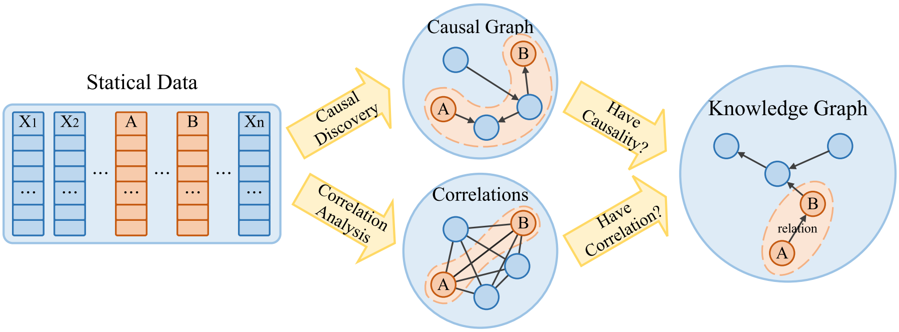


Fig. 2 General architecture of the data-based KG improvement. Both causal discovery and correlation analysis method are leveraged to uncover latent relationships between two variables, and help for knowledge graph improvement

the relationship strength. In this causal graph, different patterns of paths between variables are identified. Each pattern matches a specific type of relationship in the KG. Using these patterns, the relationship strength is calculated using a function that considers both causality and decreasing correlation. The general architecture of the proposed method is illustrated in Fig. 2. The main contributions of this paper are outlined as follows:

1. The paper introduces a new approach that uses data directly to improve KGs, eliminating the need for summarizing human knowledge. This makes the process more objective.
2. A new algorithm is suggested for enhancing KGs that is driven by data. This algorithm uses causality and correlation from the data, considering both the structure of causality and how correlation decreases. It estimates how reliable each relationship in the KG is creating a unique connection between data and knowledge.
3. Extensive experiments show that this new data-based method for improving KGs is significantly better than current models based on learning or expert, in both accuracy and stability. This improvement is further strengthened by including external data.

The remainder of this paper is organized as follows: Section 2 provides the necessary background of the research. The proposed methodology is comprehensively detailed in Sect. 3.

Experimental results and their analysis are presented in Sect. 4. Finally, the paper concludes with Sect. 5.

2 Related work

2.1 Knowledge graph improvement

Despite our best efforts to build them accurately, knowledge graphs can still have issues with quality [19]. This is due to a variety of factors, including limitations in the sources of data, difficulties in managing big data, the constantly changing nature of information, or the natural bias that comes with expert evaluations [6, 20, 21]. For example, automated knowledge graph construction methods may misinterpret text context, leading to errors. If a text states that aspirin is used by cardiovascular disease patients, including those with high blood pressure, an algorithm might wrongly conclude that aspirin reduces blood pressure, thereby introducing inaccuracies into the knowledge graph. All these factors can lead to errors and inconsistencies in KGs [22]. Many methods have been developed to improve the quality of KGs by finding and fixing errors within KGs [23, 24].

Several meticulously crafted cross-domain KGs have spurred researchers to explore methods reliant on external knowledge, such as distance supervision. This technique leverages external databases to annotate relations within a KG [25]. However, its effectiveness is contingent upon the availability of a high-quality knowledge base, a requirement that may not be met in certain specialized domains [21, 26].

Alternatively, some researchers have sought to identify inaccuracies by learning from the graph structure itself [10, 27]. This entails training a model to comprehend the KG's intrinsic structure and semantics, enabling it to ascertain the veracity of information. Yet, this strategy falters in the presence of a noisy KG, characterized by abundant incorrect or irrelevant data [28].

Expert evaluation is the most accurate methods. However, it is time-consuming and can be biased, since it depends on the individual expert's opinions and understanding, especially when knowledge must be verified using numerical data [9]. The effectiveness of this process largely depends on the expert's ability to generalize from data, which could vary widely. There is a pressing need for a method that effectively bridges the gap between data and knowledge, enabling better use of data to improve KGs.

2.2 Correlation analysis from data

Correlation, a key statistical metric, quantifies the degree to which two variables are interrelated and is widely used in various fields of data analysis [29]. The popularity of correlation in data analysis stems from its simplicity and ease of interpretation, making it an indispensable tool across multiple disciplines. While correlation provides insights into potential relationships between variables, it is crucial to note that correlation does not imply causation.

To quantitatively calculate the strength of correlation, popular methods include the Pearson correlation coefficient for continuous data and the χ^2 and G^2 tests for categorical variables, based on tests of conditional independence. These techniques not only broaden the scope of correlation analysis beyond continuous data, but also provide valuable insights in areas ranging from statistics to the social sciences.

2.3 Causal structure learning and causal inference

Delving deeper, causality unveils a more intricate and nuanced type of relationship between variables, transcending the superficial connections highlighted by correlation [30]. Causality, conceptualized as the generic link between an effect and its cause, is a pervasive phenomenon in observed data [31, 32]. A variety of algorithms, grounded in constraint-based, score-based, and functional causal models (FCMs), have been developed to learn causal structure from data [33].

Constraint-based methods, such as the PC algorithm, initiate the process with a complete undirected graph, progressively pruning edges to uncover causal relations [34]. In contrast, score-based models, epitomized by the GES architecture, adopt an iterative edge-adding approach to construct the causal graph [35]. Another score-based model proposed recently is MINOBSx [36], which proposes a precise fractional method that targets the directed acyclic graph for searching the optimal score under the current arrangement. FCM-based methods, exemplified by LiNGAM, deduce the function of each node based on its direct causes and the unobserved disturbances, leading to the construction of a causal graph [37]. These causal structure learning methods provide a robust framework for our proposed KG correction method, helping to uncover the underlying causal structures in data.

Knowledge extraction from causal graphs hinges on three foundational assumptions: the Stable Unit Treatment Value Assumption (SUTVA), ignorability, and positivity [38]. Sample re-weighting techniques, addressing selection bias, and matching methods, estimating the counterfactuals and mitigating bias from confounders, are pivotal in this context [39, 40]. Additionally, tree-based strategies, utilizing decision tree learning models such as CART, play a significant role in inference processes [41].

3 Methodology

3.1 Problem statement

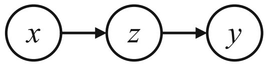
A knowledge graph is a directed heterogeneous graph $\mathcal{K} = (v, r)$ whose nodes v are entities and edges r are relations between entities. KG $\mathcal{K}$ consists of L triples, denoted as $\mathcal{K} = \{k_1, k_2, \ldots, k_L\}$. Each triple consists of a head entity h_i, a tail entity t_i, and a relation r_i, i.e., $k_i = (h_i, r_i, t_i)$. The goal is to detect and delete all the incorrect triples in $\mathcal{K}$ and to obtain the corrected KG $\mathcal{K}^*$.

Statistical data is represented as a matrix $D = X_{M\times N}$ consisting of M rows and N columns. X_i is the ith row of X, which denotes the ith sample. X^j is the jth column of X, which denotes the jth variable, corresponding to the jth entity in the KG.

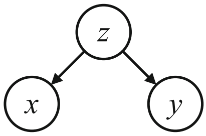
A causal graph is a directed acyclic graph $\mathcal{G}$ where each node represents a variable, and the edge represents the causality from one variable to another.

Assumption 1 In this paper, each entity in KG v^j corresponds to a variable in data matrix X^j. In other words, the data matrix can be expressed in two forms: $X_{M\times N} = [X^1, X^2, \ldots, X^N]$ and $X_{M\times N} = [v_1, v_2, \ldots, v_N]$.

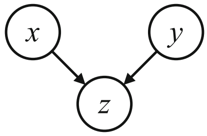
Assumption 2 Triples in KG are "relation-like," which means relation with obviously negative meaning will not appear in the KG, such as "has no effect on" or "has no relationship with."



(a) A chain structure, where x and y are independent iif z is conditioned on.



(b) A fork structure, where x and y are independent iif z is conditioned on.



(c) A collider structure, where x and y are independent iif z is not conditioned on.

Fig. 3 Three typical causal structures

3.2 Knowledge graph improvement based on integration of causal structure and correlation

3.2.1 Causal structures and causal paths

In a causal graph, there are three typical structures for conditional independence: chain, fork and collider (Fig. 3). In chain and fork structure, x and y are independent if and only if the center node z is conditioned on. However, in collider structure, x and y are independent if and only if the collider node z is not conditioned on. In other words, in collider structure, if the center node is a free variable, the side nodes are independent.

A path is a sequence of directed edges. In a causal graph, there are three types of paths: (1) collider path: there is a collider structure like $x_1 \rightarrow x_2 \leftarrow x_3$ in a path (Fig. 4a), (2) fork path: the path is not a collider path and there is a fork structure like $x_1 \leftarrow x_2 \rightarrow x_3$ in the path (Fig. 4b), and (3) sequential path: all edges in the path are in the same direction (Fig. 4c). (4) The rest situation between two variables is no path (Fig. 4d).

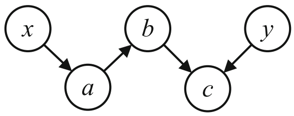
3.2.2 Conditional independence test

The conditional independence test (CI test) is a statistical way for testing the conditional independence of two variables x and y. As a basic technique for CI test, G^2-test [42] is usually used to test the independence between two discrete variables [43].

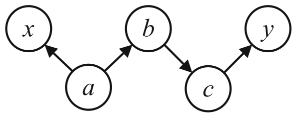
The significance level $G^2(h, t)$ of G^2-test could be obtained as follows:

$$G^2(h, t) = 2 \sum_{a=1}^{|\{X^h\}|} \sum_{b=1}^{|\{X^t\}|} O(a, b) \ln \left\{ \frac{O(a, b)}{H(a, b)} \right\} \tag{1}$$

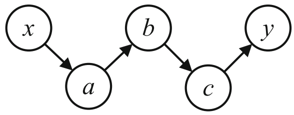
where $|\{X^h\}$ and $|\{X^t\}|$ denote the number of possible values of X^h and X^t. $O(a, b)$ is the number of samples in which the property X_h valued as $X^j(a)$ and the property X_t valued



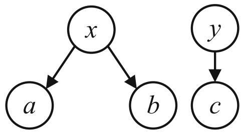
(a) The collider path is a path with a collider structure. There is a collider path between x and y, so there is no causality or correlation between x and y.



(b) The fork path, absent of collider structures but with a fork structure between x and y, results in a correlation whose strength decays with path extension.



(c) A sequential path between x and y implies a decaying correlation as the path extends and full causality



(d) There is no path between x and y, both causality and correlation between them are nonexistent.

Fig. 4 Causality and correlation in four path patterns

as $X^h(b)$. $H(a, b)$ represents the expected number of X^h valued as $X^h(a)$ and X^t valued as $X^t(b)$ when X^h and X^t are independent. The O and H are expressed as follows:

$$O(a, b) = |\{l|X_l^h = X^h(a), X_l^t = X^t(b)\}| \tag{2}$$

$$H(a, b) = \frac{|\{l|X_l^h = X^h(a)\}| \cdot |\{l|X_l^t = X^t(b)\}|}{M} \tag{3}$$

where M is the number of samples. The two terms on the molecule represent the number of samples in which h is valued as $X^h(a)$ and t as $X^t(b)$. Then, the correlation $\mathcal{C}$ between X^h and X^t can be organized by checking the cut-off value table of the G^2 distribution $\alpha(\cdot)$:

$$\mathcal{C}(h, t) = \alpha(G^2(h, t), (K_h - 1) \times (K_t - 1)) \tag{4}$$

3.2.3 Knowledge graph correction from statistical analysis

Statistical analysis is used to guide KG correlation via confirming the relation of two connected entities in a KG. In a causal graph, there could be several paths between two variables, constituting a path set P. The final relation strength between two variables is represented by the most significant one of all path relations, i.e.,

$$\mathcal{R}(h, t) = \max_{p \in P} \mathcal{PR}_p(h, t) \tag{5}$$

A threshold R_0 is used to judge whether the two variables really have a relation. Relation under R_0 would be deleted from the KG. Since different patterns of edge directions in a path may lead to different path relations, it is necessary to analyze these patterns separately.

Algorithm 1 Data-based knowledge graph improvement

Require: Knowledge graph $\mathcal{K}$, data matrix $X_{M \times N}$, relation threshold R_0, damping factor β, correlation preference γ.
Ensure: Corrected knowledge graph $\mathcal{K}^*$

Discrete every continuous variables in X
Construct causal graph $\mathcal{G}$ from data matrix X
$\mathcal{K}^* \leftarrow \mathcal{K}$ ▷ $\mathcal{K}^*$ is the optimized KG
for each $k = (h, r, t)$ in $\mathcal{K}$ **do**
  $\mathcal{R}(h, t) \leftarrow 0$
  Calculate correlation degree $\mathcal{C}(h, t)$ ▷ Equation 1-4.
  Find the path set P between h and t in $\mathcal{G}$
  for each p in P **do**
    Judge the type of path p
    Calculate path correlation value $\mathcal{PR}_p(h, t)$ ▷ Equation 6-8.
    if $\mathcal{PR}_p(h, t) > \mathcal{R}(h, t)$ **then**
      $\mathcal{R}(h, t) \leftarrow \mathcal{PR}_p(h, t)$ ▷ use path of the greatest value
    end if
  end for
  if $\mathcal{R}(h, t) \leq R_0$ ▷ r is of low quality **then**
    $\mathcal{K}^* \leftarrow \mathcal{K}^* - \{k\}$ ▷ delete k from $\mathcal{K}$
  end if
end for
Return $\mathcal{K}^*$

- *Correlation in fork path*
  If a fork path $p(x, y)$ is observed between x and y, causality could not be determined, while correlation still exists. Correlation tends to be weaker as the path extends. So an attenuation mechanism is introduced in the path to represent this decay. The relation in a fork path $\mathcal{PR}$ is estimated as:

$$\mathcal{PR}_p(x, y) = \mathcal{C}(x, y)e^{-\beta(|p|-1)} \tag{6}$$

  where p is a fork path, and $|p|$ is the path length of p. β is a damping factor that controls the attenuation rate with the path length, which is a parameter. The correlation can be calculated by Eqs. 1–4.
- *Causality in sequential path*
  A sequential path $p(x, y)$ directing from x to y strongly confirms the fact that x is the cause of y in the KG. The relation between x and y in a sequential path consists of causality and correlation. Since causality is a strong format of correlation, in this case, x and y have full causality, and the correlation between them attenuates as the path extends. So the relation in a sequential path $\mathcal{PR}$ is estimated as:

$$\mathcal{PR}_p(x, y) = \mathcal{C}(x, y)(\gamma e^{-\beta(|p|-1)} + (1 - \gamma)) \tag{7}$$

  where p is a sequential path. γ is the preference of correlation, which is a parameter to control the importance of the two parts of the relation.
- *Non-relation judgment*
  If there is no sequential path or fork path between x and y, x and y have no relation. Specifically, there are two situations: (1) there is no path between x and y (Fig. 4d). (2) there are only collider paths between x and y (Fig. 4a).
  Since the collider blocks the relation between entities, the relation between entities in a collider path is estimated to be 0:

$$\mathcal{PR}_p(x, y) = 0 \tag{8}$$

Table 1 Details of the knowledge graphs

Dataset	#E	#R	Research domain
Cancer	5	4	Smoking and cancer
Asia	8	8	Reasons causing dyspnea
Child	20	25	Childhood diseases
Insurance	27	52	Car insurance cost
Water	32	66	Waste water treatment control
Mildew	35	46	Winter wheat mildew control
Alarm	37	46	ALARM monitoring system
Barley	48	84	Production of beer

#E and #R denote the number of entities and relations in each dataset, respectively

where p is a collider path.

The comprehensive method is concluded in Algorithm 1. The algorithm first discrete each variable and constructs a causal graph using causal discovery algorithms. Then, for each triple in KG, the algorithm finds the set of paths between the head and tail entity and calculates the relation for each path. Finally, the algorithm selects the largest path strength as the final relation value and deletes the triple with low relation value.

4 Experimental results

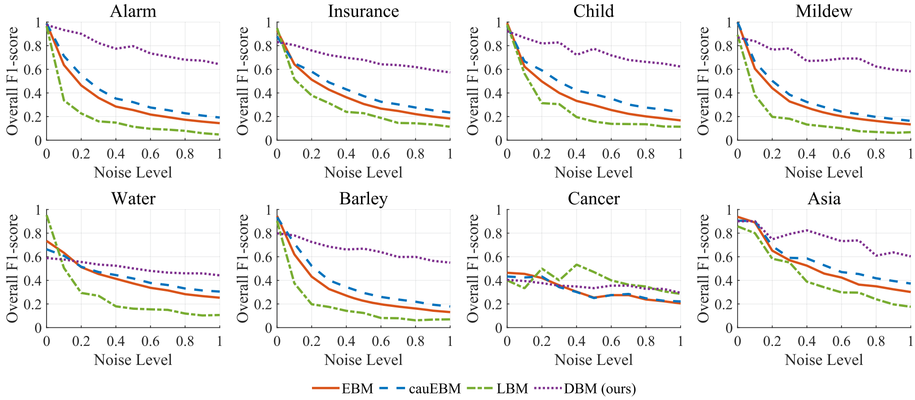
4.1 Experimental settings

In this study, the proposed method is evaluated using eight distinct knowledge graphs from various fields.[1] Each knowledge graph is meticulously constructed by domain experts corresponding to their respective fields, ensuring high quality. All relations within these KGs are of a single type, specifically labeled as "cause." Details of the number of entities and relations, along with the research domain for each KG, are presented in Table 1. Data guiding the KG improvement is generated based on the joint probabilities inherent in the Bayesian networks. We employ the MINOBSx algorithm [36] for causal graph discovery from the generated data. The strength of correlation between variables is quantified using the G^2-test [42].

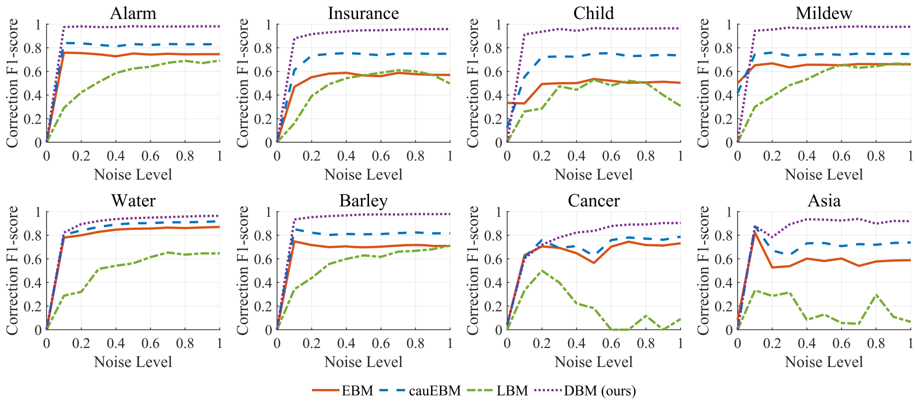
Our approach is juxtaposed with two comparison models: a learning-based model (LBM) and an expert-based model (EBM). The LBM employs the TransE algorithm [10] to learn vector representations of entities. The EBM mimics the expert knowledge acquisition process, employing a rudimentary approach for data-derived knowledge acquisition. Specifically, the method "EBM" refers to the model that designates entity pairs with a correlation surpassing a specific threshold as knowledge. The variant "cauEBM" integrates causality by treating entity pairs whose average of correlation and causality (1 if present) exceeds the threshold as a knowledge unit.

For method testing, a manually constructed knowledge graph serves as the ground truth, with random noise edges introduced into the true KG as erroneous knowledge. The noise

[1] Resources can be obtained from https://www.bnlearn.com/bnrepository/.



(a) Performance of Overall F1-score



(b) Performance of Correction F1-score

Fig. 5 F1-scores of four methods in eight datasets, under the condition of different noise level

level indicates the ratio of noisy edges added to the KG to all unconnected edges. The ability to detect and correct this noise knowledge is indicative of the performance of the methods.

4.2 Evaluation metrics

Performance metrics mainly include precision, recall, F1-score, and the structural Hamming distance (SHD).

The SHD denotes the number of knowledge that must be adjusted to transition from the evaluated knowledge graph to the ground truth. These adjustments address three types of errors: "missing," "extra," and "reverse." Specifically, "missing" and "extra" correspond to the absent and superfluous erroneous knowledge, respectively, in comparison with the ground truth. Meanwhile, "reverse" errors indicate instances where the knowledge graph incorrectly inverts the direction of the actual relationship. A lower SHD indicates superior performance as it suggests that the resultant knowledge graph can be rectified to align with the ground truth with fewer modifications.

Metrics labeled as “Correction” represent the evaluation outcomes for erroneous knowledge induced by noise, while those labeled as “Overall” refer to the evaluation outcomes based on the ground truth.

4.3 Results and analysis

4.3.1 Performance comparison under different noise levels

To showcase the efficacy of the method proposed in this paper, we conducted experiments on eight datasets, with results displayed in Fig. 5.

As can be observed, the proposed method, referred to as DBM, consistently outperforms in the majority of parameter spaces. Notably, a significant performance gap often exists between DBM and its closest competitor. Regarding the overall F1 metric, DBM’s performance inevitably experiences a gradual decline as noise levels increase. However, it still retains the best performance amidst these conditions. In terms of correction F1, DBM’s performance progressively improves, underlining its superior capability to identify and rectify erroneous knowledge.

4.3.2 Fine comparison of different methods under different metrics

Table 2 provides a snapshot of experimental results with a noise level of 0.2, offering a more detailed perspective on various performance metrics across different methods.

From the results, it is evident that the LBM consistently exhibits significant shortcomings either in precision or recall. For the overall F1, LBM’s precision is exceptionally low, suggesting a tendency to inaccurately classify correct knowledge. In terms of correction F1, LBM’s recall is quite low, indicating a difficulty in identifying nonexistent knowledge. High recall, denoting superior error detection capability, is particularly crucial in areas with minimal tolerance for errors.

The EBM exhibits slightly better performance, given that it relies on objective data that can accurately represent knowledge. The performance of cauEBM surpasses that of EBM, suggesting that knowledge extracted with causality as a reference is more appropriate for knowledge scenarios than relying solely on correlation. This can be reasonably attributed to the fact that correlations exhibited by entities under certain conditions may not accurately represent their true relationships, and thus are not considered knowledge by human standards.

However, cauEBM’s design merely represents a rudimentary use of causality and correlation. Although it approaches or even exceeds DBM’s performance on some datasets, its overall performance still falls short of DBM.

Additionally, we also compared the SHD with detailed sub-metrics of various methods, as depicted in Fig. 6.

Under this metric, DBM continues to exhibit the best performance. To be more specific, for the “reverse” sub-metric, the outcomes across all methods are comparable as these techniques do not differentiate functions or modules of relationship asymmetry. For the “missing” and “extra” sub-metrics, DBM records the lowest scores. This suggests that in terms of the output, DBM’s knowledge graph is the closest to the “ideal” representation of the ground truth.

Table 2 Performances of all methods in each dataset (noise level = 0.2)

Dataset	Category	Metric	EBM	cauEBM	LBM	DBM	Dataset	Category	Metric	EBM	cauEBM	LBM	DBM
Asia	Overall	Precision	51.76	59.82	43.75	59.41	Cancer	Overall	Precision	60.14	75.00	50.00	80.70
		Recall	88.54	83.33	87.50	86.46			Recall	39.58	35.42	50.00	66.67
		F1-score	65.17	69.07	58.33	**70.27**			F1-score	42.03	43.25	50.00	**71.26**
	Correction	Precision	83.39	83.13	66.67	85.86		Correction	Precision	62.84	63.02	50.00	74.03
		Recall	39.39	57.58	18.18	56.82			Recall	85.42	97.92	50.00	83.33
		F1-score	52.71	67.04	28.57	**68.03**			F1-score	70.54	75.84	50.00	**77.30**
Child	Overall	Precision	33.38	43.50	20.78	69.53	Mildew	Overall	Precision	28.28	33.65	12.18	57.40
		Recall	98.33	93.67	64.00	87.33			Recall	100.0	99.64	52.17	77.36
		F1-score	49.74	59.12	31.37	**77.14**			F1-score	44.07	50.27	19.75	**65.72**
	Correction	Precision	98.71	96.85	60.87	95.49		Correction	Precision	100.0	99.89	73.81	95.29
		Recall	33.67	58.33	18.67	87.00			Recall	50.21	61.28	26.38	88.76
		F1-score	49.34	72.26	28.57	**90.98**			F1-score	66.80	75.88	38.87	**91.89**
Insurance	Overall	Precision	36.48	46.96	26.32	54.14	Water	Overall	Precision	45.90	54.06	19.89	55.39
		Recall	87.02	78.53	67.31	77.24			Recall	58.08	49.75	56.06	44.95
		F1-score	51.23	58.09	37.84	**62.98**			F1-score	51.18	**51.53**	29.37	49.50
	Correction	Precision	89.41	88.75	68.52	89.67		Correction	Precision	84.08	83.11	59.15	82.15
		Recall	40.74	63.77	27.41	73.58			Recall	76.26	85.17	21.99	87.44
		F1-score	55.13	73.44	39.15	**80.45**			F1-score	79.94	84.08	32.06	**84.69**
Alarm	Overall	Precision	30.39	38.47	13.70	67.19	Barley	Overall	Precision	28.42	37.32	12.25	59.00
		Recall	97.83	97.65	65.22	95.47			Recall	89.79	87.80	51.19	68.85
		F1-score	46.37	55.12	22.64	**78.81**			F1-score	43.15	52.21	19.77	**63.48**
	Correction	Precision	99.38	99.44	82.42	99.15		Correction	Precision	96.71	96.88	76.84	93.92
		Recall	60.92	72.51	28.41	91.79			Recall	57.02	71.51	30.63	90.94
		F1-score	75.53	83.82	42.25	**95.32**			F1-score	71.68	82.17	43.80	**92.40**

bold values refers the best result of each group

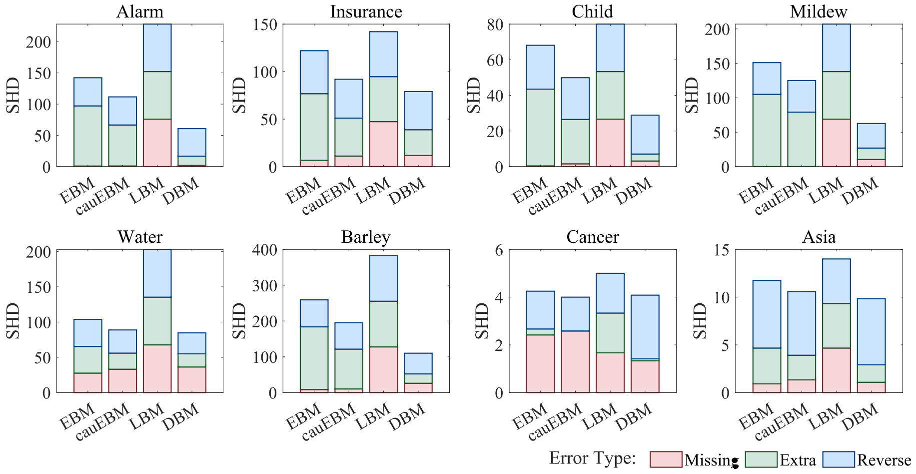
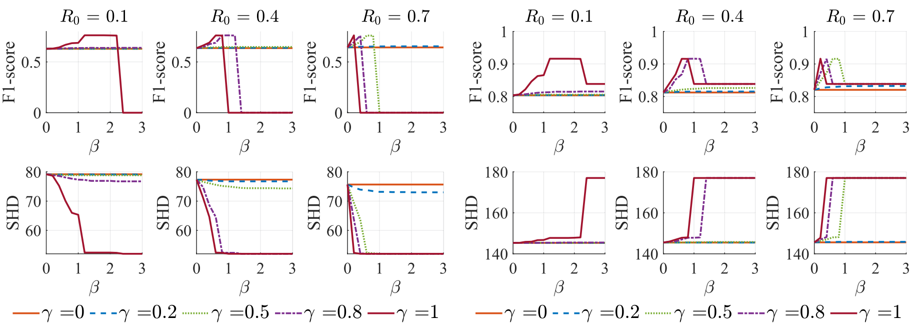


Fig. 6 Overall structural Hamming distance (SHD) and the corresponding counts for missing, extra, and reversed relations for four different methods (lower values indicate better performance)



(a) Overall F1-score and SHD (b) Correction F1-score and SHD

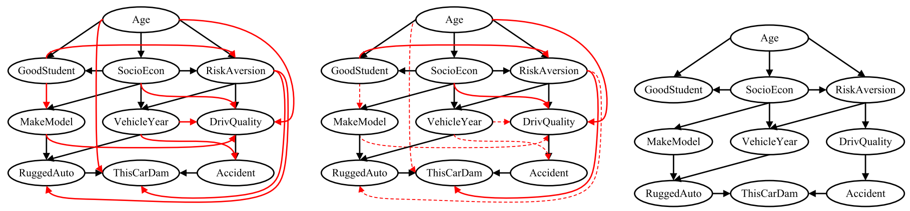
Fig. 7 Experimental results on insurance dataset under varied parameters. Each figure illustrates the range of F1-scores or SHD obtained with different values of γ and β, and R_0 varies across each column of figure. All experiments were conducted at a fixed noise level of 0.2

4.3.3 Performance influenced by parameters

The experimental results, which are contingent on various parameters, are presented in Fig. 7. The subsequent analysis uncovers several key trends and patterns:

An increase in β results in a fluctuating response in the F1-scores. Some F1-scores exhibit an initial upward trend, followed by a decline, achieving a maximum within a specific range. Conversely, others remain constant, lingering at lower values. The presence or absence of a peak in the F1-score is dictated by the values of R_0 and γ. A larger R_0 and a higher γ not only facilitate the occurrence of this peak, but also confine its β range. Such a phenomenon could be ascribed to the fact that an excessively small γ leads to a profound bias toward model correlation, thereby impeding the potential benefits of data causality.

Simultaneously, as β escalates, the overall SHD experiences a substantial reduction under specific parameter conditions. However, upon reaching a certain threshold, the correction SHD associated with these parameters sees a pronounced increase. As a result, the perfor-



(a) The noisy KG. (b) KG improved by the DBM. (c) The ground truth KG.

Fig. 8 Visualization of the experimental results on the "Insurance" KG, where the relation names "cause" is omitted. The correct relations are represented by black arrows, incorrect relations by red arrows, and red dashed lines indicate the incorrect connections removed by the proposed DBM

mance of β within these two boundary values is deemed more efficient. Interestingly, this range and the parameters associated with it mirror the peak region's range corresponding to the F1-score.

4.3.4 Effectiveness in applications

To delve deeper into the application scenarios and actual efficiency of the proposed method, we provide a more intuitive display and analysis through a case study.

"Insurance" is a knowledge graph in the car insurance industry, encompassing various causally related knowledge areas specific to this field [44]. This KG includes 27 entities and 52 relations. For ease of demonstration, a sub-KG containing 10 entities and 14 relations is randomly selected for detailed study. The names, meanings, and values of the entities included in this KG are introduced as follows:

- *GoodStudent* Whether the policyholder is a good student. Values: True, False.
- *Age* The policyholder's age group. Values: Adolescent, Adult, Senior.
- *SocioEcon* The policyholder's socioeconomic status. Values: Prole, Middle, UpperMiddle, Wealthy.
- *RiskAversion* The policyholder's risk aversion level. Values: Psychopath, Adventurous, Normal, Cautious.
- *VehicleYear* The recency of the policyholder's vehicle. Values: Current, Older.
- *ThisCarDam* The extent of damage to the vehicle. Values: None, Mild, Moderate, Severe.
- *RuggedAuto* The ruggedness of the vehicle. Values: EggShell, Football, Tank.
- *Accident* The severity of the policyholder's accident, including the damage to all included cars. Values: None, Mild, Moderate, Severe.
- *MakeModel* The make and model of the policyholder's vehicle. Values: SportsCar, Economy, FamilySedan, Luxury, SuperLuxury.
- *DrivQuality* The quality of the policyholder's driving. Values: Poor, Normal, Excellent.

In this KG, ten noisy relations are created by setting the noise level at 0.1. Subsequently, the proposed DBM method is applied to eliminate the noisy relations and optimize this noisy KG. The results are illustrated in Fig. 8.

As observed, most of the noisy relations were identified and removed by our algorithm, including some relations that are difficult to verify directly, demonstrating the effectiveness of the proposed method. The noisy relations that were not removed are mostly those that have indirect causal relations. For example, the unremoved noisy relation "Age → DriveQuality" contains an causal relation in the data. However, in this scenario, their causal relation is indirect but occurs through “RiskAversion” as an intermediary variable. Thus, this relation was not successfully removed by the proposed method. Similar reasoning applies to relations such as “SocioEcon → DrivQuality” and “RiskAversion → ThisCarDam.” These instances indicate a potential direction for improvement in future work.

5 Conclusions

In this work, we introduced an novel data-centric methodology designed to improve the quality of knowledge graphs, particularly for professional sectors that are characterized by an abundance of data. The proposed algorithm meticulously extracts causality and correlation from the data, utilizing these insights to assess and affirm the integrity of existing relations within the KG. This process not only narrows the chasm between raw data and refined knowledge, but also results in enhancements in both accuracy and stability. The proposed approach has been thoroughly tested through extensive experiments, showing that our model for KG refinement outperforms existing learning-based alternatives in both accuracy and stability.

In the era of large language models (LLMs), this study provides new insights. On the one hand, our research indicates that the vast amounts of numeric data commonly available contain a wealth of knowledge, making them a potential source for the LLM training. On the other hand, the extensive knowledge embedded in LLMs can also serve as a vital resource for enhancing knowledge graphs [45]. The aspiration of this study is to combine the cutting-edge developments in natural language processing and machine learning with data-based strategy, thereby enhancing the objectivity and usability of KGs, especially for professional applications.

Author Contributions Derui Lyu conceptualized the main ideas and led the project. Xiangyu Wang and Lyuzhou Chen contributed significantly to the analysis and interpretation of data. Taiyu Ban and Kaiming Xiao were involved in drafting the manuscript and revising it critically for important intellectual content. Lihua Liu provided final approval of the version to be published and agreed to all aspects of the work.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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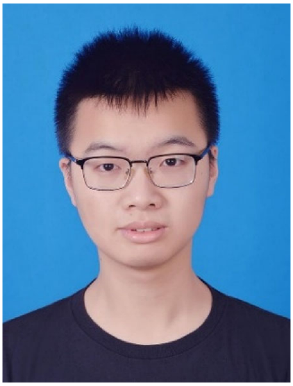

Derui Lyu received the B.Sc. degree in intelligence science and technology from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2021. He is currently pursuing the Ph.D. degree in computer science and technology with the School of Computer Science and Technology, University of Science and Technology of China (USTC), Hefei, China. His current research interests include machine learning and knowledge-driven image recognition.

Xiangyu Wang received the B.Sc. degree from the Donghua University, Shanghai, China, and the Ph.D. degree in data science from the University of Science and Technology of China (USTC). He is currently an associate researcher with the School of Computer Science and Technology, USTC. His research interests include knowledge engineering and machine learning.

Lyuzhou Chen received the B.Sc. degree in mathematics and applied mathematics from the University of Science and Technology of China, Hefei, China, in 2020, where he is currently pursuing the Ph.D. degree in computer science and technology with the School of Artificial Intelligence and Data Science. His current research interests include knowledge engineering and causal learning.

Taiyu Ban received the B.Sc. degree in computer science and technology from the University of Science and Technology of China, Hefei, China, in 2020, where he is currently pursuing the Ph.D. degree in computer science and technology with the School of Computer Science and Technology. His current research interests include data-driven machine learning and knowledge engineering.

Kaiming Xiao PhD, is a Lecturer of Laboratory for Big Data and Decision in National University of Defense Technology. His research interests include big data, data-driven intelligent decision, game theory. He has published more than 20 research papers in leading conference and journals including IJCAI, INFOCOM, AAAI, AASMAS, ECAI.

Lihua Liu received the B.S. and Ph.D. degrees in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2004 and 2012, respectively. She is currently an Associate Professor in the Laboratory for Big Data and Decision, NUDT. Her research interests include target recognition, data fusion, data engineering, and intelligent decision.